



Uncertainty Estimation in Landmark Localization Based on Gaussian Heatmaps

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Abstract. In landmark localization, due to ambiguities in defining their exact position, landmark annotations may suffer from both large inter- and intra-observer variabilities, which result in uncertain annotations. Therefore, predicting a single coordinate for a landmark is not sufficient for modeling the distribution of possible landmark locations. We propose to learn the Gaussian covariances of target heatmaps, such that covariances for pointed heatmaps correspond to more certain landmarks and covariances for flat heatmaps to more uncertain or ambiguous landmarks. By fitting Gaussian functions to the predicted heatmaps, our method is able to obtain landmark location distributions, which model location uncertainties. We show on a dataset of left hand radiographs and on a dataset of lateral cephalograms that the predicted uncertainties correlate with the landmark error, as well as inter-observer variabilities.

Keywords: Landmark localization · Uncertainty estimation

1 Introduction

Anatomical landmark localization is an important topic in medical image analysis, e.g., as a preprocessing step for segmentation [1, 9], registration [11, 19], as well as for deriving surgical or diagnostic measures like the location of bones in the hip [2], the curvature of the spine [20], and the misalignment of teeth or the jaw [21]. Unfortunately, anatomical landmarks can be difficult to define unambiguously, especially for landmarks that do not lie on distinct anatomical structures like the tip of the incisor, but on smooth edges like the tip of the chin. Such ambiguous landmarks are specifically difficult to annotate, which

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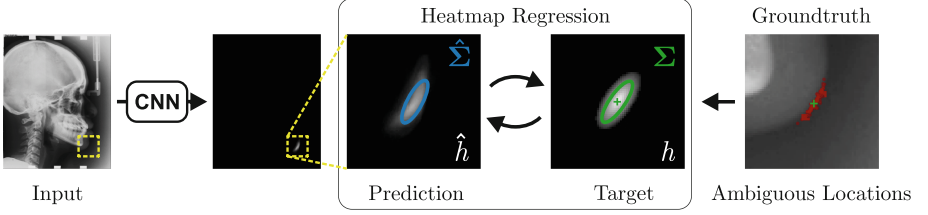


Fig. 1. Schematic representation of our proposed method to model annotation ambiguities. Although ambiguities are not observable from a single groundtruth annotation (green cross), they are present in the entirety of the annotated images (red dots). We model landmark ambiguities with a Gaussian function with covariance Σ that is learned during training and represents a *dataset-based uncertainty*. The predicted heatmap $\hat{h}$ is fitted with covariance $\hat{\Sigma}$ to represent the predicted landmark locations as a distribution, modeling an *image-based uncertainty*. (Color figure online)

may lead not only to a large inter-observer variability between different annotators, but also to a large intra-observer variability that is dependent on the daily constitution of a single annotator. Due to this uncertainty, a machine learning predictor that has been trained on potentially ambiguous annotations should model landmarks not only as single locations, but rather as distributions over possible locations. Predicting such uncertainty measures helps interpreting the output of machine learning programs [6], which is especially useful in the medical imaging domain [14, 22, 23] where explainability is crucial [8].

Methods in anatomical landmark localization predominantly use machine learning. While random forests have been successfully applied [5, 13, 18], recently, convolutional neural networks (CNNs) outperformed other techniques, especially when doing heatmap regression [2, 15–17, 24]. In these works, heatmaps have only been used for predicting single locations, e.g., as the maximum response, and not for obtaining distributions of possible landmark locations.

We hypothesize that the heatmap regression framework may be used to model landmark uncertainties (see Fig. 1). We adapt the framework to learn the optimal shape of target heatmaps, which models an uncertainty based on annotation ambiguities in the training dataset. Furthermore, by fitting a Gaussian function to the predicted heatmap during inference, we do not only predict the most probable landmark location but its distribution, which models a prediction uncertainty. Evaluation on datasets of left hand radiographs and lateral cephalograms shows that the uncertainties correlate with the magnitude and direction of the average localization error, as well as the inter-observer variabilities.

2 Heatmap Regression for Dataset-Based Uncertainty

We perform anatomical landmark localization with CNNs using heatmap regression [2, 15–17, 24]. For N target landmarks, the CNN with parameters $\mathbf{w}$ predicts N heatmaps $\hat{h}_i(\mathbf{x}; \mathbf{w})$, while each heatmap corresponds to a single landmark L_i

with $i = \{1, \dots, N\}$. A target heatmap is represented as an image of a two-dimensional Gaussian function $h_i(\mathbf{x}; \mathbf{\Sigma}_i) : \mathbb{R}^2 \rightarrow \mathbb{R}$, i.e.,

$$h_i(\mathbf{x}; \mathbf{\Sigma}_i) = \frac{\gamma}{\sqrt{(2\pi)^2 \det \mathbf{\Sigma}_i}} \exp \left(-\frac{1}{2} (\mathbf{x} - \mathbf{x}_i)^T \mathbf{\Sigma}_i^{-1} (\mathbf{x} - \mathbf{x}_i) \right). \quad (1)$$

The mean of the Gaussian is set to target landmark's coordinate $\mathbf{x}_i \in \mathbb{R}^2$, while the shape of the Gaussian is defined by the covariance matrix $\mathbf{\Sigma}_i$. A scaling factor γ is used to avoid numerical instabilities during training.

Differently to other heatmap regression methods, which represent the covariance of the Gaussian with only a single parameter σ_i , we allow anisotropic Gaussian functions with a general covariance matrix $\mathbf{\Sigma}_i$. This way, we do not model isotropic, axis-aligned Gaussian functions with varying extents but anisotropic Gaussian functions that have different extents in different directions. The covariance matrix $\mathbf{\Sigma}_i$ may be decomposed into

$$\mathbf{\Sigma}_i = \mathbf{R}_i \mathbf{\Sigma}_i^* \mathbf{R}_i^T, \text{ with } \mathbf{R}_i = \begin{bmatrix} \cos \theta_i & -\sin \theta_i \\ \sin \theta_i & \cos \theta_i \end{bmatrix} \text{ and } \mathbf{\Sigma}_i^* = \begin{bmatrix} \sigma_i^{\text{maj}} & 0 \\ 0 & \sigma_i^{\text{min}} \end{bmatrix}, \quad (2)$$

where θ_i represents the rotation of the Gaussian function's major axis, and σ_i^{maj} and σ_i^{min} represent its extent in major and minor axis, respectively.

The loss function for simultaneously regressing N heatmaps is defined as

$$\min_{\mathbf{w}, \mathbf{\Sigma}_1, \dots, \mathbf{\Sigma}_N} \sum_{i=1}^N \sum_{\mathbf{x}} \|\hat{h}_i(\mathbf{x}; \mathbf{w}) - h_i(\mathbf{x}; \mathbf{\Sigma}_i)\|_2^2 + \alpha \sum_{i=1}^N \sigma_i^{\text{maj}} \sigma_i^{\text{min}}. \quad (3)$$

The first term in (3) minimizes the differences between the predicted heatmaps $\hat{h}_i$ and the target heatmaps h_i for all landmarks L_i . Since we also treat the covariance parameters of the Gaussian function (θ_i , σ_i^{maj} , and σ_i^{min}) as unknowns, we enable learning them in addition to the network parameters $\mathbf{w}$. Unfortunately, minimizing the difference between $\hat{h}_i$ and h_i could lead to the trivial solution $\sigma_i^{\text{maj}} \sigma_i^{\text{min}} \rightarrow \infty$ with $h_i \approx 0$. To avoid that $\sigma_i^{\text{maj}} \sigma_i^{\text{min}} \rightarrow \infty$, the second term in (3) penalizes $\sigma_i^{\text{maj}} \sigma_i^{\text{min}}$ with factor α .

Analyzing (3) in more detail, not only the predicted heatmaps $\hat{h}_i$ aim to be close to the target heatmaps h_i but also the target heatmaps h_i aim to be close to the predicted heatmaps $\hat{h}_i$. For each landmark L_i , $\hat{h}_i$ and h_i receive feedback from each other during training (see Fig. 1). While the network parameters $\mathbf{w}$ are updated to better model the shape of the target heatmap $\hat{h}_i$, at the same time, the covariance $\mathbf{\Sigma}_i$ parameters of each target heatmap h_i (θ_i , σ_i^{maj} , and σ_i^{min}) are updated to better model the shape of the predicted heatmap $\hat{h}_i$. Note that $\mathbf{\Sigma}_i$ is learned only from single annotations per image, but from all annotated training images, thus, modeling the ambiguities of landmark annotations for the whole training dataset and representing a *dataset-based uncertainty*.

3 Heatmap Fitting for Image-Based Uncertainty

As the output of heatmap regression networks are N heatmaps that correspond to the N landmarks L_i , the landmark's coordinate $\hat{\mathbf{x}}_i$ needs to be obtained from

the predicted heatmaps $\hat{h}_i$. Previous work often uses the coordinate of the maximum response or the center of mass of the predicted heatmaps as the landmark’s coordinate. However, this represents only the most probable landmark location, but not the whole distribution of possible landmark locations.

As the target heatmaps are modeled as Gaussian functions (see (1)), we expect also the predicted heatmaps to be a Gaussian function. Hence, we capture the distribution of possible landmark locations by fitting Gaussian functions to the predicted heatmaps $\hat{h}$. We use a robust least squares curve fitting method [3] to fit the Gaussian function (see (1)) around the maximum of the predicted heatmaps $\hat{h}_i$ and obtain the fitted Gaussian parameters $\hat{\mathbf{x}}_i$ and $\hat{\Sigma}_i$ with $i = \{1, \dots, N\}$. As the fitted Gaussian function models the distribution of possible locations, the parameters represent a directional *image-based uncertainty* of the landmark location during inference.

4 Experimental Setup

Networks: Our proposed method to learn anisotropic Gaussian heatmaps can be combined with any image-to-image based network architecture. In this work, we use the SpatialConfiguration-Net (SCN [16]), due to its state-of-the-art performance and publicly available code. We implement learning of Gaussian functions with arbitrary covariance matrices. We use the SCN with the default parameters trained for 40,000 iterations. We use training data augmentation, i.e., random intensity shift and scale, translation, scaling, rotation, and elastic deformation. We use L_2 weight regularization with a factor of $\lambda = 0.001$. In (1) we set $\gamma = 100$; in (3), we initialize $\sigma_i^{\text{maj}} = \sigma_i^{\text{min}} = 3$, $\theta_i = 0$, and set $\alpha = 5$.

We perform an ablation study on different strategies of using Gaussians. First, we distinguish whether the target Gaussian heatmaps in (2) are isotropic, i.e., $\sigma^{\text{maj}} = \sigma^{\text{min}}$, or anisotropic during training. Second, we distinguish how the location is obtained from the predicted heatmaps during inference. This can be done by either taking the maximum [15–17, 24], or, as we are proposing, by fitting Gaussians. We compare networks using isotropic Gaussians as target and taking the maximum (σ -target, max), fitting isotropic (σ -target, $\hat{\sigma}$ -fit) and anisotropic Gaussians (σ -target, $\hat{\Sigma}$ -fit), as well as networks using anisotropic Gaussians for target only (Σ -target, $\hat{\sigma}$ -fit), and for both target and fitting (Σ -target, $\hat{\Sigma}$ -fit).

Datasets: We evaluate our proposed method on two publicly available datasets for anatomical landmark localization. The first dataset consists of 895 radiographs of left hands of the Digital Hand Atlas Database System [7, 25] with 37 annotated landmarks per image. Each image has been annotated by one of three experts. The second dataset consists of 400 lateral cephalograms with 19 annotated landmarks per image. This dataset has been used for the ISBI 2015 Cephalometric X-ray Image Analysis Challenge [21], in which each image has been annotated by both a senior and a junior radiologist. As every image has been annotated by two radiologists, we are able to measure the inter-observer variability in this dataset and can compare it with the predicted uncertainty.

Metrics: For evaluating the landmark localization performance, we use the point-to-point error PE, i.e., the Euclidean distance between groundtruth $\mathbf{x}$ and prediction $\hat{\mathbf{x}}$. We also calculate the number of outliers $\#O_r$ for radius r , which is defined as the number of landmarks with a larger PE of r mm. The outlier ratio OR_r is the percentage of outliers to all predicted landmarks; the success detection rate is defined as $SDR_r = 100 - OR_r$.

Table 1. Quantitative localization results. Top: Results for the cross-validation (CV) on the hand radiographs. Bottom: Results on the test sets (using the mean annotation of junior and senior), as well as the CV (using the junior’s annotation) for the lateral cephalograms. Best values are marked in bold.

<i>hand</i>	Method	PE (in mm)		#O _r (OR _r in %)		
		median	mean $\pm$ SD	$r = 2$ mm	$r = 4$ mm	$r = 10$ mm
<i>CV</i>	Lindner et al. [12]	0.64	0.85 ± 1.01	2094 (6.32%)	347 (1.05%)	20 (0.06%)
	Urschler et al. [18]	0.51	0.80 ± 0.93	2586 (7.81%)	510 (1.54%)	18 (0.05%)
	Payer et al. [16]	0.43	0.66 ± 0.74	1659 (5.01%)	241 (0.73%)	3 (0.01%)
	(σ -target, max)	0.40	0.62 ± 0.72	1507 (4.55%)	263 (0.79%)	4 (0.01%)
	(σ -target, $\hat{\sigma}$ -fit)	0.40	0.61 ± 0.67	1381 (4.17%)	185 (0.56%)	3 (0.01%)
	(σ -target, $\hat{\Sigma}$ -fit)	0.40	0.61 ± 0.66	1354 (4.09%)	161 (0.49%)	3 (0.01%)
	(Σ -target, $\hat{\sigma}$ -fit)	0.39	0.61 ± 0.68	1404 (4.24%)	169 (0.51%)	5 (0.02%)
	(Σ -target, $\hat{\Sigma}$ -fit)	0.39	0.61 ± 0.67	1349 (4.07%)	151 (0.46%)	4 (0.01%)
<i>ceph.</i>	Method	PE (in mm)	SDR _r in %			
		mean $\pm$ SD	$r = 2$ mm	$r = 2.5$ mm	$r = 3$ mm	$r = 4$ mm
<i>Test1</i>	Ibragimov et al. [10]	$1.67 \pm \text{n/a}$	71.72%	77.40%	81.93%	88.04%
	Lindner et al. [12]	$1.84 \pm \text{n/a}$	73.68%	80.21%	85.19%	91.47%
	Zhong et al. [26]	1.12 ± 0.88	86.91%	91.82%	94.88%	97.90%
	Chen et al. [4]	$1.17 \pm \text{n/a}$	86.67%	92.67%	95.54%	98.53%
	(Σ -target, $\hat{\Sigma}$ -fit)	1.07 ± 1.02	87.37%	91.86%	94.81%	97.79%
<i>Test2</i>	Ibragimov et al. [10]	$1.92 \pm \text{n/a}$	62.74%	70.47%	76.53%	85.11%
	Lindner et al. [12]	$2.14 \pm \text{n/a}$	66.11%	72.00%	77.63%	87.42%
	Zhong et al. [26]	1.42 ± 0.84	76.00%	82.90%	88.74%	94.32%
	Chen et al. [4]	$1.48 \pm \text{n/a}$	75.05%	82.84%	88.53%	95.05%
	(Σ -target, $\hat{\Sigma}$ -fit)	1.38 ± 1.33	75.11%	82.53%	88.26%	94.58%
<i>CV jun.</i>	Lindner et al. [13]	$1.20 \pm \text{n/a}$	84.70%	89.38%	92.62%	96.30%
	Zhong et al. [26]	1.22 ± 2.45	86.06%	90.84%	94.04%	97.28%
	(σ -target, max)	1.02 ± 1.13	88.72%	92.55%	94.83%	97.42%
	(σ -target, $\hat{\sigma}$ -fit)	1.00 ± 1.08	89.64%	93.32%	95.66%	97.91%
	(σ -target, $\hat{\Sigma}$ -fit)	1.00 ± 1.07	89.78%	93.58%	95.80%	97.95%
	(Σ -target, $\hat{\sigma}$ -fit)	0.99 ± 1.08	89.63%	93.55%	95.66%	97.83%
	(Σ -target, $\hat{\Sigma}$ -fit)	0.99 ± 1.07	89.76%	93.74%	95.83%	97.82%

5 Results and Discussion

Comparison with Other Methods: While the main focus of our paper lies on evaluating the uncertainty measures, we validate our method by comparing to the state-of-the-art in terms of landmark localization error in Table 1. The reported metrics and values are taken from the respective publications.

On a three-fold cross-validation (CV) of the dataset of left hand radiographs, our proposed method outperforms all previous results. When comparing the different fitting strategies, we can see that defining the landmark coordinate as the maximum value of the heatmap generated from the network with isotropic target heatmaps (σ -target, max) leads to the smallest improvements as compared to the state-of-the-art. When using either isotropic or anisotropic Gaussians during training and fitting isotropic ones during inference, the results increase slightly (σ -target, $\hat{\sigma}$ -fit and Σ -target, $\hat{\sigma}$ -fit). Finally, fitting anisotropic Gaussians leads to better results (σ -target, $\hat{\Sigma}$ -fit and Σ -target, $\hat{\Sigma}$ -fit), while also using anisotropic Gaussians during training leads to the overall best results (Σ -target, $\hat{\Sigma}$ -fit).

We compare to other methods on both test sets *Test1* and *Test2* of the lateral cephalograms, where we trained and evaluated networks with the average coordinate from both radiologist annotations. Here, the methods using CNNs have similar results, while our method is the most accurate with the smallest PE. In terms of SDR, our method performs in-line with other methods, while Chen et al. [4] perform slightly better in *Test1* and Zhong et al. [26] in *Test2*, respectively.

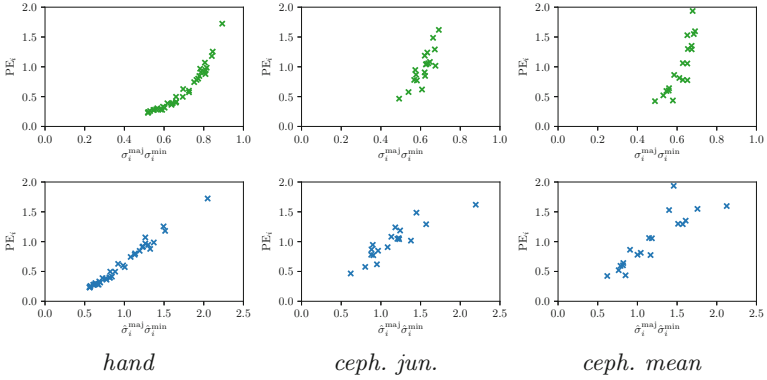


Fig. 2. Correlation of the mean learned target Gaussian sizes (top) and fitted Gaussian sizes (bottom) of all landmarks L_i to the mean localization error PE_i for all images of the CV of hand radiographs (left), cephalograms with junior annotations (center), and mean of senior and junior (right) in mm.

As we show in later experiments, there exist systematic shifts for some landmarks in *Test1* and *Test2* as compared to the annotations of the training data. To mitigate this, we follow previous work [13, 26] and perform a four-fold CV experiment trained and evaluated on the junior annotations only. Similar to the hand radiographs, learning and fitting anisotropic Gaussian functions (Σ -target, $\hat{\Sigma}$ -fit) leads to the best results, outperforming all other methods.

Uncertainty Estimation of the Landmark Distributions: Our proposed method models a *dataset-based uncertainty* during training as parameters of the target Gaussian heatmaps (see Sect. 2), as well as an *image-based uncertainty* during inference by fitting a Gaussian function to the predicted heatmaps

(see Sect. 3). For evaluating whether the parameters of the target and fitted Gaussians represent uncertainty measures, we compare the sizes of the Gaussians for each landmark L_i with the average PE_i in Fig. 2. We evaluate the CV of the hand dataset, as well as CVs of the cephalograms trained on either the junior annotations only, or the mean of both junior and senior. We can see that both the target and fitted heatmap parameters correlate with the PE, thus representing valid uncertainty measures. We want to highlight that the *dataset-based uncertainty* is obtained during network training without requiring validation images. Thus, it can make use of all annotations of the training dataset and allows identification of difficult to annotate landmarks, even with only a few annotated images.

Moreover, our proposed method is also able to model directional uncertainties, which we visualize in Fig. 3 for selected landmarks (for all landmarks, see the supplementary materials). Here, while both target and predicted heatmaps

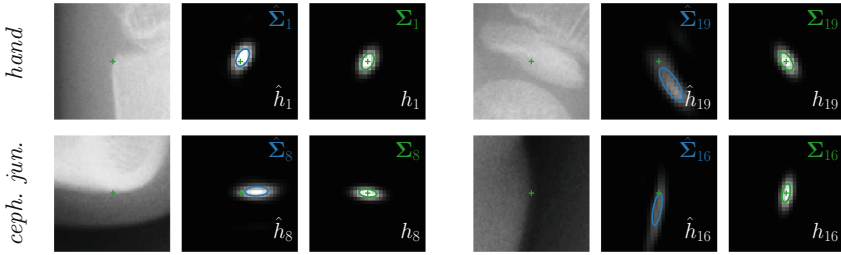


Fig. 3. Input (left), heatmap predictions (middle) and target heatmaps (right) for selected landmarks. Images show 15×15 mm patches around groundtruth annotation (green cross). The target and predicted heatmaps are superimposed with ellipses representing the target and fitted Gaussian parameters, respectively. (Color figure online)

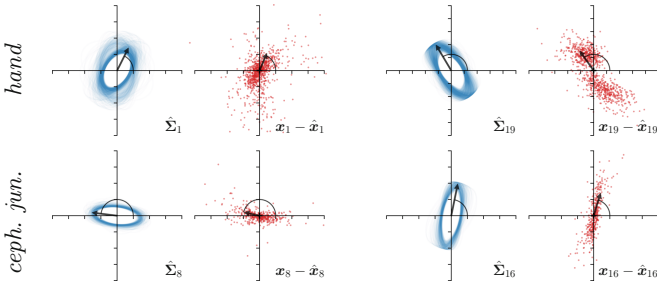


Fig. 4. Gaussian functions fitted to the predicted heatmaps (left) and offsets of the groundtruth coordinate to the predictions (right) for selected landmarks. For all images of the datasets, the fitted Gaussian parameters are shown as blue ellipses, the offsets as red dots. The average major axis of the fitted Gaussian parameter, as well as the major axis of a Gaussian distribution fitted to the offsets, are shown as black arrows. Each tick on the axis lines marks one mm.

are smaller for landmarks on distinct corners as compared to landmarks on edges, the heatmaps are also rotated in direction of the underlying edges, as this is the main source for ambiguities. Interestingly, for landmark L_{19} of the hand dataset and landmark L_{16} of the cephalograms, the predicted heatmap has a larger extent as compared to the target heatmap. Landmark L_{19} of the hand dataset lies on the base of the proximal phalanx of the thumb, which is often wrongly annotated as the head of the metacarpal. For landmark L_{16} of the cephalogram dataset, the predicted heatmaps have a larger extent, as this landmark is difficult to annotate unambiguously, which can also be seen in a large inter-observer variability as shown in later evaluations. Our method is able to detect such ambiguous landmarks by predicting a larger uncertainty expressed as more disperse distributions of landmark locations.

To show that the fitted heatmap parameters $\hat{\Sigma}_i$ represent the same distribution as the expected prediction-groundtruth offsets $\mathbf{x}_i - \hat{\mathbf{x}}_i$, we visualize all fitted heatmaps and offsets for all images of the dataset in Fig. 4 for some example landmarks. Here we can see that the fitted heatmap parameters have the same distribution as the expected groundtruth-prediction offsets. For landmark L_{19} of the hand radiograph dataset, we can see two clusters of offsets, showing well the misannotations as described previously. Overall, these results further show that the fitted parameters $\hat{\Sigma}_i$ model an image-based landmark location uncertainty.

Uncertainty Estimation for Modeling the Inter-observer Variability:

As a final experiment, we evaluate how well our proposed method models the inter-observer variability on the dataset of cephalograms. In Fig. 5, we show the predicted fitted Gaussian parameters, the groundtruth-prediction offsets, as well as groundtruth-senior and groundtruth-junior offsets for the CV trained on the mean of both junior and senior annotations. The plots show that our proposed method models well the inter-observer variability, leading to fitted Gaussian parameters $\hat{\Sigma}_i$ that are similar to the expected offsets $\mathbf{x}_i - \hat{\mathbf{x}}_i$. On these plots we can also see that there may exist a systematic offset between junior (green) and senior (yellow) annotations, leading to two distinct point clusters as compared to their mean $\mathbf{x}_i$. For landmark L_{16} , which is difficult to annotate unambiguously, the disagreement of junior and senior is very large on the training

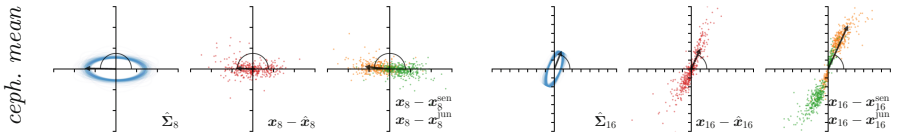


Fig. 5. Gaussian parameters fitted to the predicted heatmaps (left), offsets of the groundtruth coordinate to the predictions (middle), and groundtruth-senior and groundtruth-junior offsets (right) for selected landmarks. The visualization is the same as in Fig. 4, while the offsets of senior and junior annotation to their mean are shown as green and orange dots, respectively. (Color figure online)

dataset. Interestingly, in *Test2* their disagreement reduced from more 5 mm to ≈ 2 mm. This systematic shift of the landmark groundtruth locations for L_{16} can explain the reduced performance of all compared methods in Table 1 for *Test2*. In conclusion, these results show that the fitted heatmap parameters $\hat{\Sigma}_i$ model also the landmark location uncertainty in terms of inter-observer variability.

6 Conclusion

We have shown that CNNs using heatmap regression may be used for predicting location uncertainties. By learning Gaussian parameters of the target heatmaps and fitting Gaussian functions to the predicted heatmaps, our method is able to predict distributions of possible landmark locations. While showing state-of-the-art performance in two datasets, the predicted landmark distributions correlate with the landmark localization error as well as inter-observer variabilities. In future work, we plan to extend our method to three dimensional data.

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